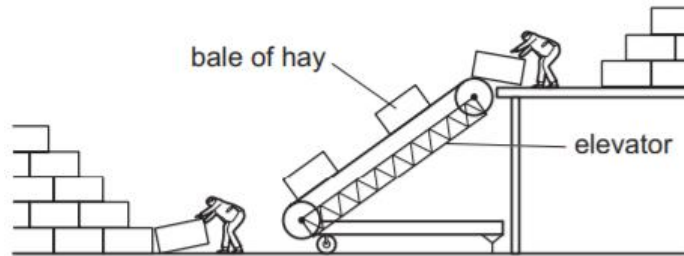


9

Two farmers use an electrically powered elevator to lift bales of hay. All the bales of hay have the same mass.



As sunset approaches, they increase the speed of the elevator so that more bales are lifted up in a given time.

How does this affect the work done in lifting each bale and the useful power output of the elevator?

		work done in lifting each bale	useful output power of the elevator	
	A	increases	decreases	
	B	increases	increases	
	C	no change	decreases	
	D	no change	increases	

Ans: D

each bale: same increase in gravitational potential energy
more bale lifted therefore higher useful output power

10

Two identical blocks connected by light rigid rods are pivoted at one end to move in a circular